



In-situ fabricated hierarchical nanostructure on titanium alloy as highly stable and durable super-lubricated surface for anti-biofouling in marine engineering

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ABSTRACT

Titanium alloys, with superior corrosion resistance, have been widely used in marine engineering. However, they are biocompatible and attractive to biofouling organisms. In this work, based on the micro-arc oxidation and hydrothermal treatment, a Slippery Liquid-Infused Porous and Prickly Surface (SLIPPS) was in-situ fabricated on titanium alloy (TA2) via a cost-efficient and facile process. The hierarchical nanostructure consisting of micro/nano-pores and nano-prickles was infused by lubricant to form a super-lubricated surface. The micro/nano-pores can offer sufficient space to store the lubricant, and the tiny prickles on/inside the pores can provide the firm mechanical interlock with the lubricant. These unique features can ensure the long-lasting storage and conservation of the lubricant, which endows the excellent stability and durability. The SLIPPS remains well lubricated after a series of tests with extreme conditions, such as the high-rate centrifugation, scouring test and long-time immersion. As an inert barrier, this stable and durable lubricant layer also significantly improves the corrosion resistance. Compared with pristine TA2 and super-hydrophobic samples, the SLIPPS possesses an impressive anti-biofouling performance. After being immersed for 31 days with two typical biofouling organisms (*Chlorella* and *Phaeodactylum tricornutum*), the coverage percentages are only 0.03% and 0.06% on SLIPPS, respectively, while for the pristine TA2, the corresponding values are 53.3% and 23.2%, respectively. Therefore, this in-situ fabricated, low-cost and super-lubricated coating on the titanium alloy surface can effectively, stably and durably prevent from biofouling and corrosion in seawater, which may pave the way for the large-scale applications of titanium alloys in marine engineering.

1. Introduction

Due to the excellent anti-corrosion capability and comprehensive mechanical properties, titanium alloys are called 'marine metals' and widely applied in marine facilities [1], such as deep-sea submersibles [2,3], ship components [4], hydrocarbon extraction devices [5], heat exchangers and condenser tubes [6–8] and so on. On the other hand, the biofouling issue is raised by the undesired attachment and accumulation of microorganisms, arthropods and mollusks on the surfaces of materials, which causes tremendous economic losses and threatens the service safety of all the marine facilities [9–11]. Even worse, titanium alloys are highly biocompatible [6] and suffer from the biofouling problem more seriously than other structural materials, which

profoundly impairs their long-term safety and functionalities. In other words, the biofouling issue is a gargantuan obstacle to employing titanium alloys in the large-scale marine applications [8].

Recently, many scientists and researchers have dedicated their efforts to alleviating or solving the biofouling problem on the surfaces of different structural materials. Traditional toxic coatings are widely adopted to repel or kill the attached organisms on various materials, including titanium alloys [12–16], steel [17], aluminum alloy [18,19], etc., but the active ingredients in the coatings are usually heavy metal ions. Although these coatings can remain effective for years before being re-painted, they are harmful to all marine creatures, and the contaminated seafood and ocean environment can eventually influence our daily life and health. This vital disadvantage is opposite to the growing

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consensus of the environmental conservation around the world, and many countries have taken actions to ban this kind of toxic anti-biofouling coatings [10,20]. As an alternative anti-biofouling approach based on well-designed micro/nano surface textures, the super-hydrophobic modification is an emerging technology to significantly reduce the surface energy of a variety of materials, like titanium alloys [21], zinc [22], steel [23], polyvinylidene fluoride (PVDF) [24], etc., and effectively trap a layer of air under the water as an isolation barrier against the biofouling organisms. However, when immersed in water for a period, the trapped air will eventually detach or dissolve in water [25], and the anti-biofouling performance of the super-hydrophobic surface will remarkably deteriorate [26–29]. In addition, neither the traditional toxic coatings nor super-hydrophobic surfaces cannot prevent biomacromolecules, such as proteins, polysaccharides and dead microorganisms, from attaching and accumulating on the materials. These organic substances can provide abundant nutrition and attract organisms [30,31], which will considerably facilitate and expedite the biofouling process.

Inspired by the slippery inner surface of the *Nepenthes* pitcher cup, Aizenberg et al. reported two nano/micro-structured surfaces with different materials (epoxy-resin and Teflon, respectively) for anti-biofouling applications, which are named as Slippery Liquid-Infused Porous Surface (SLIPS) [32]. Due to the infused lubricant in nano/micro-porous structure, a SLIPS normally has a very low adhesion force with biomacromolecules, microorganisms and other marine creatures, which makes it hardly be attached by any substance conducive to biofouling [32]. As a highly effective and environment-friendly method, the SLIPS attracts the attention of researchers worldwide and has been adopted to modify the surfaces of various materials [11,33–38]. Xiao et al. developed a SLIPS based on micrometer-scale pores prepared by the UV-initiated free-radical polymerization of butyl methacrylate – ethylene dimethacrylate, and this SLIPS exhibits a remarkable repelling effect on both spores and cyprids [36]. Li and his colleagues demonstrated that the SLIPS on poly(butyl methacrylate-co-ethylene dimethacrylate) surface has an excellent bacteria-resistance in a BM2 mineral medium for at least up to seven days [37]. Alexander et al. reported a SLIPS on Teflon membranes significantly reducing the attachment areas of *Pseudomonas aeruginosa* (99.6%), *Staphylococcus aureus* (97.2%) and *Escherichia coli* (96%) over 7 days under the static and physiological real flow conditions [39]. Additionally, Ware et al. proved that a SLIPS on Teflon can not only prevent *Pseudoalteromonas* spp. bacteria from attaching but also reduce up to 99% of their reproduction number, when only $0.9 \mu\text{L}\cdot\text{cm}^{-2}$ of silicone oil is infused [38].

Although the SLIPSs have exhibited the outstanding performance of anti-biofouling on diverse materials, the relevant studies on titanium alloys are very limited. Most of the existing fabrication methods are for non-metallic materials [11,34–38,40–45], while only few ones for metals [33,46,47], let alone titanium alloys [48]. Qiu et al. [48] report a SLIPS on titanium by electrochemically coating a silver surface, and this SLIPS can effectively keep its anti-biofouling performance for 14 days. However, the complicated process with the expensive silver coating may hinder its potential applications on large-scale surfaces of marine facilities. Therefore, a new fabrication method of a stable and durable slippery surface for titanium alloys should be proposed and developed. In fact, the main mechanism of the SLIPS to prevent from biofouling is the slippery surface, for example, the lubricant surface, which can disrupt the adhesion attempts of the microorganisms [39]. Hence, to boost the stability and durability of the SLIPS, the key point is how to preserve the infused lubricant in the open marine environment. On the other hand, the actual marine environment is complicated, and especially, the water flow can be frequent, ceaseless and in any direction. The water flow can generate not only the in-plane shear force to wash the lubricant away but also the out-of-plane normal force, resulting from the pressure gradient or eddies, to suck the lubricant out. Although the current SLIPSs provide the storage for the lubricant via micro/nano-pores [38,44,49], the mechanical interlock between the SLIPS and lubricant is still weak,

which cannot stop the loss of the lubricant in a long time period.

In this work, a Slippery Liquid-Infused Porous and Prickly Surface (SLIPPS) was in-situ fabricated on a titanium alloy (TA2) via the micro-arc oxidation technology and chemical modification, which is a cost-efficient and facile process. The SLIPPS is a hierarchical nanostructure consisting of two parts to improve its stability and durability for anti-biofouling: 1) micro/nano-pores provide the storage for the lubricant; 2) nano-prickles inside/on the porous surface firmly retain the lubricant. All the micro/nano structures were directly made of titanium alloy, and no extra metal or other material is needed. The enhanced capability of the SLIPPS to preserve the lubricant is demonstrated by a series of tests with extreme conditions, like the high-rate centrifugation, scouring test and long-time immersion. *Chlorella* and *Phaeodactylum tricornutum* were employed as the typical biofouling organisms to prove the excellent long-term anti-biofouling performance of the SLIPPS. Additionally, the stable and durable lubricant layer as an inert barrier can also significantly enhance the corrosion resistance of the SLIPPS. This low-cost, stable, durable and super-lubricated surface, with its remarkable biofouling resistance and corrosion resistance, could elongate the shortest stave of the Liebig's barrel for titanium alloys used in marine engineering and greatly expand their potential in large-scale applications.

2. Experimental section

2.1. Materials

The titanium alloy (TA2 plates with a thickness of 2 mm) were fabricated by Hunan GoldSky Titanium Industry Technology Co., Ltd. (Hunan, China), and the dimensions of the samples are $10 \text{ mm} \times 10 \text{ mm} \times 2 \text{ mm}$. Sodium silicate, sodium hydroxide, potassium fluoride and trichloro (1H, 1H, 2H, 2H-perfluorooctyl) silane were purchased from Tianjin Damao Chemical Reagent Factory (Tianjin, China), Shanghai Aladdin Biochemical Technology Co., Ltd. (Shanghai, China), Guangzhou Chemical Reagent Factory (Guangdong, China) and AI LAN Chemical Technology Co., Ltd. (Shanghai, China), respectively. Sodium metaphosphate and glycerol were obtained from Shanghai Macklin biochemical technology Co., Ltd. (Shanghai, China). Perfluoropolyether (PFPE) (Krytox GPL103) was purchased from Suzhou Zhengfengyi Lubricating Material Co., Ltd. (Jiangsu, China). *Chlorella* and *Phaeodactylum tricornutum* were purchased from Zhuhai Xinrui Trading Co., Ltd. (Guangdong, China). Glutaraldehyde (2.5%) was obtained from Shanghai Yuanye Biotechnology Co., Ltd. (Shanghai, China). The deionized (DI) water was prepared by a laboratory pure water machine (EKUP-II-308, EKEA, China). The artificial seawater was prepared according to the ASTM D1141 standard.

2.2. Anti-biofouling coatings preparation

The in-situ process flow of the SLIPPS is schematized in Fig. 1. Firstly, a TA2 plate was polished with gritted silicon carbide sandpapers and rinsed with acetone, ethanol and DI water, successively. Then, a Porous Surface (PS) on the TA2 was fabricated via a Micro-Arc Oxidation (MAO) process. The MAO equipment (MAO-30KW, Jinchuangli, China) was purchased from Chengdu Jinchuangli Technology Co., Ltd. In the MAO process, a pulsed bi-polar power source with a cooling system was utilized, and the electrolyte was composed of $15 \text{ g}\cdot\text{L}^{-1}$ sodium silicate, $6 \text{ g}\cdot\text{L}^{-1}$ glycerol, $7 \text{ g}\cdot\text{L}^{-1}$ potassium fluoride, $7 \text{ g}\cdot\text{L}^{-1}$ sodium hydroxide, $2 \text{ g}\cdot\text{L}^{-1}$ sodium metaphosphate and an appropriate amount of DI water. The MAO process was conducted with the constant current mode, and the average current density is $+0.7 \text{ A}\cdot\text{cm}^{-2}$ with a 6% duty cycle and an 800 Hz frequency. After the MAO process, the PS sample was carefully cleaned with DI water and dried in air.

To achieve the hierarchical nanostructure on the TA2 surface, the PS further underwent a hydrothermal treatment to form a Porous and Prickly Surface (PPS). In the hydrothermal treatment, the PS sample was

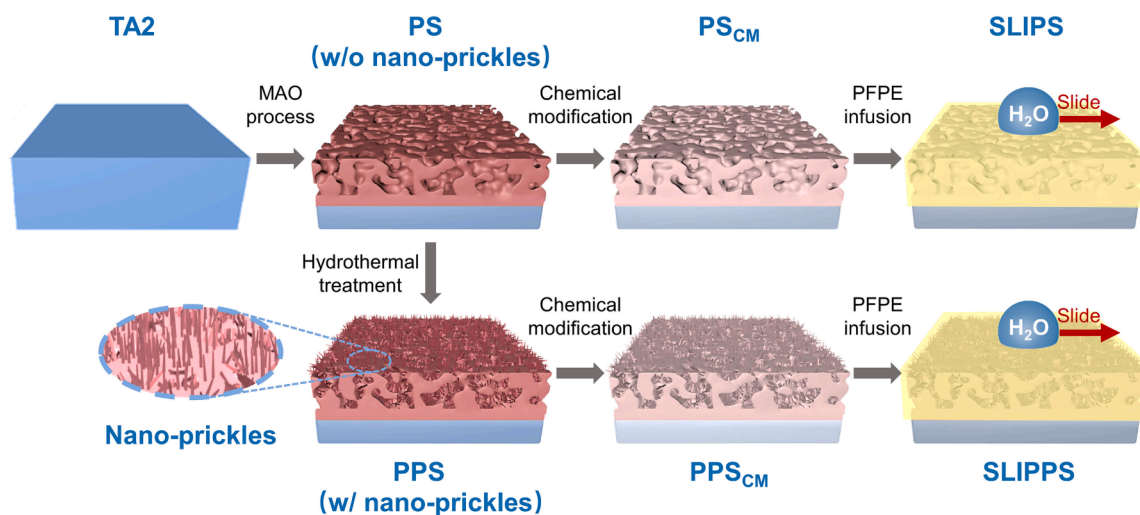


Fig. 1. Schematic illustration of the process flow for the SLIPS and SLIPPS.

immersed into a 4 wt% sodium hydroxide aqueous solution at 40 °C for 4 hrs., and the nano-prickles were formed inside/on the porous surface. To improve the affinity between the PPS and lubricant, a chemical modification of the surface was conducted via immersing the samples into a 20 mL alcohol solution of 20 μ L Trichloro (1H, 1H, 2H, 2H-perfluorooctyl) silane for 4 hrs. and subsequently baking at 100 °C for 60 min. As a control group, a PS sample without nano-prickles was also chemically treated with the same way, and the chemical modified PS and PPS samples were named as PS_{CM} and PPS_{CM}, respectively. Finally, several droplets of PFPE (~20 μ L) as the lubricant were added on the PS_{CM} and PPS_{CM} until the sample surfaces were fully covered with the PFPE to obtain SLIPS and SLIPPS, respectively. The extra PFPE were removed by keeping the samples vertically at least for 1 hr. The photos of all the fabricated samples, including pristine TA2, PS, PPS, PS_{CM}, PPS_{CM}, SLIPS and SLIPPS ones, are shown in Fig. S1.

2.3. Characterization

The surface morphology and chemical composition were examined by a Field-Emission Scanning Electron Microscope (FE-SEM, SU8010, Hitachi, Japan) equipped with an Energy Dispersive X-ray Spectrometer (EDS). The samples were sealed with epoxy resin before cross-section SEM and EDS tests. The chemical compositions of different samples were measured by an X-ray Diffraction (XRD, SmartLab, Rigaku, Japan). The Raman spectra of the samples were collected on a laser confocal Raman spectroscopy system (inVia Qontor, Renishaw, UK). The binding energies of different elements in the samples were analyzed by an X-ray Photoelectron Spectroscopy system (XPS, Nexsa, Thermo Scientific, USA). The functional groups were examined by a Fourier Transform Infrared (FTIR) spectroscopy system (Nicolet 6700, Thermo Scientific, USA). The micro-FTIR spectrum mapping tests were conducted at a FTIR spectroscope (Vertex 70, Bruker, Germany) coupled with a Vis-IR microscope (Hyperion 3000, Bruker, Germany). The Contact Angle (CA) and Sliding Angle (SA) with a water droplet of the sample surfaces were measured by a droplet shape analyzer (DSA100S, KRÜSS, Germany). The corrosion resistance data of the samples were analyzed in artificial seawater with an electrochemical workstation (Interface1010E, GAMRY, Germany), including a standard three-electrode system with a saturated calomel reference electrode, a platinum-plated counter electrode and the sample as the working electrode. The zeta potentials of the microorganism cells were measured with a nanoparticle size – zeta potential and molecular weight measurement analyzer (EliteSizer, Brookhaven, USA). The zeta potential of the SLIPPS by an electrokinetic analyzer (SurPASS3, Anton Paar, Austria).

2.4. Stability and durability test

The stability and durability of the samples were evaluated by the high-rate centrifugation, scouring and long-time immersion. In the high-rate centrifugation test, the samples were placed in the centrifuge hole, the lubricated surfaces faced outwards, and the spin rate ramped up to a set value (0–12000 rpm) on a table high-rate centrifuge (LC-LX-H185C, Lichen, China) and maintained the set value for 1 min. The high-rate spinning could generate a considerable centrifugal force to facilitate the loss of the lubricant, which could mimic the out-of-plane normal force of the seawater flow in real applications. According to the changes of the water sliding angle and sample weight before and after the above test, the capability of each sample to preserve the lubricant against the normal force could be estimated. In the scouring test, the samples were moved at various velocities from 0 to 4.8 m/min in a thermostatic water bath vibrator (SHA-BA, Langyue, China) for 15 min. The water flow in parallel with the lubricated surfaces could mimic the in-plane shear force of the seawater flow in practical. The changes of the water sliding angle and sample weight before and after this scouring test could also be used to evaluate the capability of the samples to retain the lubricant against the shear force. In the long-time immersion test, the samples were immersed in artificial seawater from 0 to 7 days, and the water CA was measured before and after the immersion to assess the effect of this test.

2.5. Anti-biofouling test

In the anti-biofouling test, the sterilized artificial seawater was used as the medium to cultivate the *Chlorella* and *Phaeodactylum tricornutum* in a light incubator (QHP-360BE, Lichen, China), and the light/dark time ratio was 12 hrs. / 12 hrs. The titanium alloy samples were placed in the media with different types of algae for 1 day, 7 days and 31 days, respectively. Then, the samples were taken out and immersed in the 2.5% glutaraldehyde to fix the biofouling microorganisms on the sample surfaces for 1 hr. Before the optical analysis, the surfaces of the samples were mildly washed with the artificial seawater and dried in the air. Finally, the coverage percentages of the biofouling microorganisms on the sample surfaces were observed by a Confocal Laser Scanning Microscopy (CLSM, LSM710 NLO, ZEISS, Germany) to estimate the anti-biofouling performance.

3. Results and discussion

3.1. Preparation of anti-biofouling coatings

To prepare the SLIPS and SLIPPS coatings on TA2, the micro-arc oxidation (MAO) was firstly used to fabricate a Porous Surface (PS). Then, for the SLIPS process, the PS was chemically modified with fluoroalkane to improve its chemical affinity with the lubricant, and the sample was named as the PS_{CM} after the chemical modification process. Finally, the PS_{CM} was coated with the lubricant (Perfluoropolyether, PFPE), and the surplus PFPE was removed to obtain the SLIPS. For the SLIPPS process, the PPS was obtained by an additional hydrothermal treatment on the PS, followed by the chemical modification to obtain the PPS_{CM} and the PFPE infusion to obtain the SLIPPS. The schematic illustration of the process flow for the SLIPS and SLIPPS is shown in Fig. 1.

3.2. Surface morphology and chemical composition of anti-biofouling coatings

Fig. 2a-c show the Scanning Electron Microscopy (SEM) images and Energy Dispersive Spectroscopy (EDS) mappings of the cross-section of the PS coating, to examine the thickness and element distribution of the coating just after the MAO process. In the SEM and EDS tests, a resin layer was used to seal the sample for the cross-section observations. In Fig. 2a-b, it can be observed that the thickness of the coating is about 10 μm , which can be hardly changed after the MAO process. Additionally, the coating is comprised of a looser outer surface layer and an inner compact layer, which is consistent with other works [50,51]. The EDS mappings in Fig. 2c clearly indicate that the coating is mainly composed of Ti, Si, P and O elements in the cross-section view, which could form titanium oxides, silicon oxides and phosphorus oxides. The titanium element comes from the TA2 substrate, while the silicon and phosphorus elements come from the electrolyte during the MAO process. Because the MAO process involves the electrochemistry and plasma chemistry, these complex reactions occur at the metal/oxide and oxide/electrolyte boundaries. Eventually, an oxide ceramic layer grows in-situ on the substrate surface [51].

In order to investigate the phase composition of the as-fabricated hierarchical nanostructure, the TA2, PS and PPS samples were characterized with the X-Ray Diffraction (XRD) and Raman spectroscopy, respectively. As shown in Fig. 2d, the XRD result of the pristine TA2 is a typical titanium pattern (JCPDS 44-1294), while both the PS and PPS consist of titanium and titanium oxide, which includes the TiO_2 rutile phase (JCPDS 21-1276) and TiO_2 anatase phase (JCPDS 21-1272) [50,52]. Fig. 2e shows the Raman spectra of the PS and PPS, and the TiO_2 rutile phase (444 cm^{-1} , 610 cm^{-1}) and TiO_2 anatase phase (149 cm^{-1} , 395 cm^{-1} , 518 cm^{-1} , 641 cm^{-1}) are observed in both the PS and PPS [53,54], which is consistent with the XRD results.

Furthermore, the contact angle (CA) measurements of the TA2, PS_{CM} and PPS_{CM} were also conducted to investigate the wettability change after the chemical modification, and the results are shown in Fig. 2f. Notably, the chemical modification process can only change the chemical status of the sample surface, but not change any morphology of it. Apparently, the pristine TA2 sample has a typical hydrophilic surface, while both the PS_{CM} and PPS_{CM} samples are hydrophobic with the CAs of 139.1° and 162.2° , respectively. Especially, the PPS_{CM} possesses a CA over 150° and exhibits the super-hydrophobicity. In theory, the wettability for a solid surface is determined by two factors: the surface energy (chemical composition) and surface roughness (micro/nano structure) [55,56]. Normally, the surface energy depends on the electronic structure of the surface material. After the chemical modification with fluoroalkane, the C-F bond formed on the both surfaces of the PS_{CM} and PPS_{CM} is a full octet of the electrons, which is difficult to form the hydrogen bond with water and results in a low surface energy [57]. In the Wenzel Equation of a rough surface [58]: $\cos\theta_r = R\cos\theta$, the θ_r is the CA of a rough surface, the θ is the CA of the corresponding smooth surface in the classic Young's Equation, and the R is the relative surface roughness. The R is equal to the ratio of a rough surface area to its corresponding smooth surface area. Since the R of a rough surface is always greater than 1 [43,59], when the CA is larger than 90° , the relative surface roughness has a magnifying effect on the CA of the solid surface. Therefore, enhanced by the higher relative surface roughness resulting from the nano-prickles, the CA of the PPS_{CM} is larger than that of the PS_{CM} .

Since the wettability for a solid surface is influenced by the relative

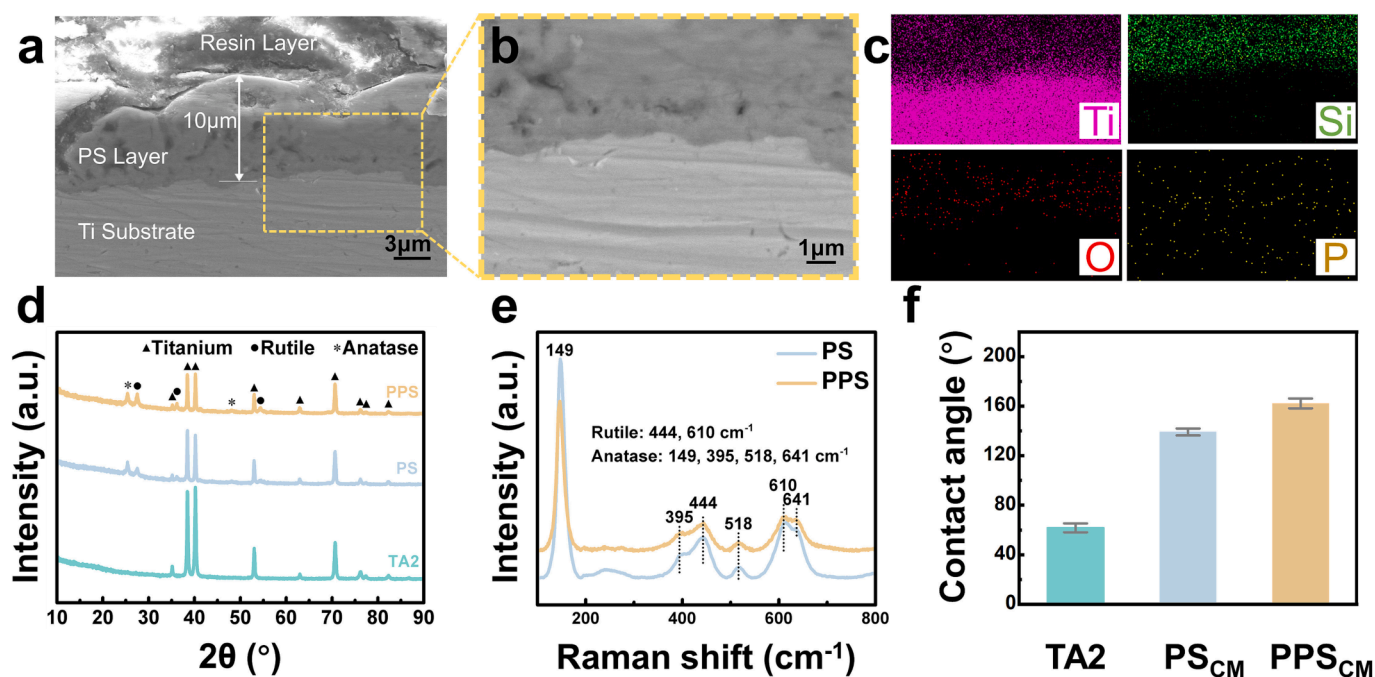


Fig. 2. (a) SEM image of the cross-section of the PS. (b) Detailed SEM image of the coating-substrate interface in (a). (c) EDS mappings of the cross-section of the PS. (d) XRD patterns and (e) Raman shift spectra of the TA2, PS and PPS. (f) CAs of the TA2, PS_{CM} and PPS_{CM} .

surface roughness, the surface morphology details of the PS and PPS samples were characterized by SEM, as shown in Fig. 3. The PS surface is composed of many micro-pores (Fig. 3a-b), and there are also micro/nano-pores on the inner wall of the big micro-pores or on the ridges between the big micro-pores, which is a typical porous foam structure [60]. However, the PPS surface is not only a typical porous foam structure, but also there are a lot of nano-prickles in-situ growing on/inside the pores (Fig. 3c-d). In fact, such a unique structure directly leads to an increase in the specific surface area and provides a better chemical affinity and physical interaction with the infused lubricant after the chemical modification process [36]. In Fig. S2, a typical SEM image of the cross-section of the PPS coating is presented. Apparently, the nano-prickles only in-situ grows on the inner wall surfaces of the pores (Pore A and Pore B in Fig. S2), while the skeleton is still a dense solid and without nano-prickles.

To further ascertain the change of the chemical composition during the fabrication process, the PPS, PPS_{CM} and SLIPPS samples were characterized by the X-ray Photoelectron Spectroscopy (XPS). As shown in Fig. 4a, the spectrum of the PPS contains Ti2p, Si2p and O1s peaks, which indicates its main elements before the chemical modification. As a contrast, after the chemical modification process, the XPS survey spectrum of PPS_{CM} displays an additional F1s peak (688.76 eV), which comes from the chemical modification compounds. For the SLIPPS, the Ti2p (459.46 eV) and Si2p (103.03 eV) peaks disappear, due to the infusion of the PFPE, which demonstrates that the PFPE lubricant has been infused into the PPS_{CM} uniformly and firmly covers the PPS_{CM}.

To analyze the influence of the lubricant infusion process with more details, the deconvoluted spectra for the C1s and O1s peaks of the PPS_{CM} and SLIPPS are displayed in Fig. 4b-e. The C1s spectrum of the PPS_{CM} is deconvoluted to the C-C (284.9 eV), C-O (286.0 eV), C-F₂ (291.5 eV), C-F₃ (293.95 eV) and CH₂-CF₂ (289.2 eV) peaks [61], which are the bonds

in the PPS_{CM} after the chemical modification process. However, in the deconvoluted C1s spectrum of the SLIPPS, the peaks of C-F₂ (291.2 eV) and C-F₃ (294.2 eV) correspond to the bonds of the PFPE [62]. On the other hand, the deconvoluted O1s spectrum of the PPS_{CM} shows the presence of three peaks located at 532.5 eV, 531.1 eV, 530.5 eV, which are attributed to the Ti-O-Si, O-H and O²⁻ peaks, respectively [63]. The presence of the Ti-O-Si covalent bond indicates that the organic compounds were successfully grafted to the sample surface during the chemical modification. However, the deconvoluted O1s spectrum of SLIPPS only displays a single peak of the C-O-F bond (536.4 eV), which is attributed to the existence of the PFPE [64]. These XPS results confirm that a PFPE layer were prepared successfully on the surface of PPS_{CM} via the lubricant infusion.

3.3. Stability and durability of anti-biofouling coatings

A stable and durable lubricant film is crucial for a super-lubricated surface, to ensure its expected performance. Based on the minimization of the total energy, to keep the infused liquid being retained in the porous surface spontaneously, the following two equations should be satisfied [32]:

$$R(\gamma_B \cos \theta_{B,r} - \gamma_A \cos \theta_{A,r}) - \gamma_{AB} > 0 \quad (1)$$

$$R(\gamma_B \cos \theta_{B,r} - \gamma_A \cos \theta_{A,r}) + \gamma_A - \gamma_B > 0 \quad (2)$$

Here, the subscript A represents the isolated or repelled liquid, like the water, while the subscript B denotes the infused liquid, like the lubricant. The γ_A and γ_B are the surface tensions of the isolated liquid and infused liquid, respectively. The γ_{AB} is the interfacial tension between the isolated liquid and infused liquid. The $\theta_{A,r}$ and $\theta_{B,r}$ are the CAs of a rough or porous surface for the isolated liquid and infused liquid, respectively. In this work, the γ_A , γ_B and γ_{AB} are 72.4 mN•m⁻¹, 17.7 mN•m⁻¹ and 53.7 mN•m⁻¹, respectively [65]. For the PPS_{CM} and

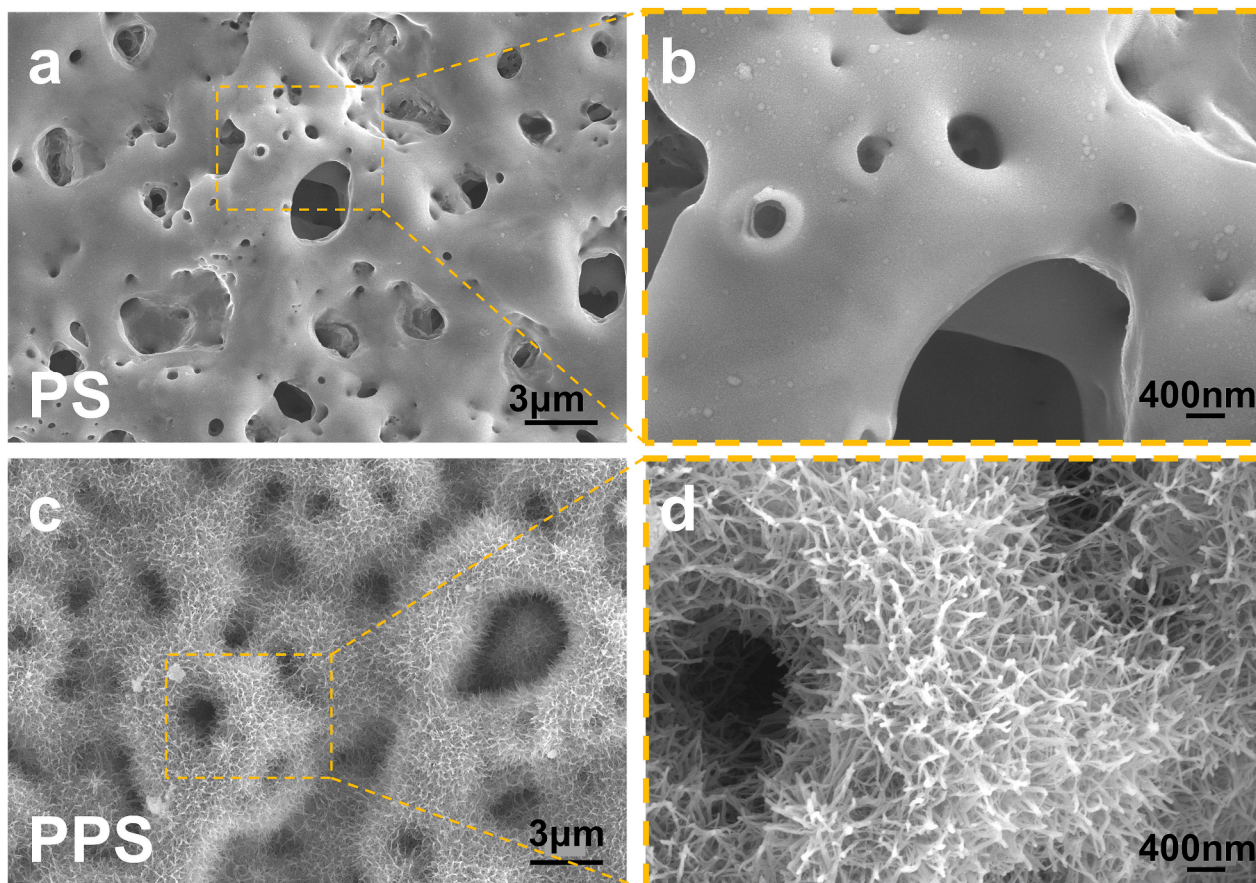


Fig. 3. SEM images of the surface morphologies of the (a-b) PS and (c-d) PPS.

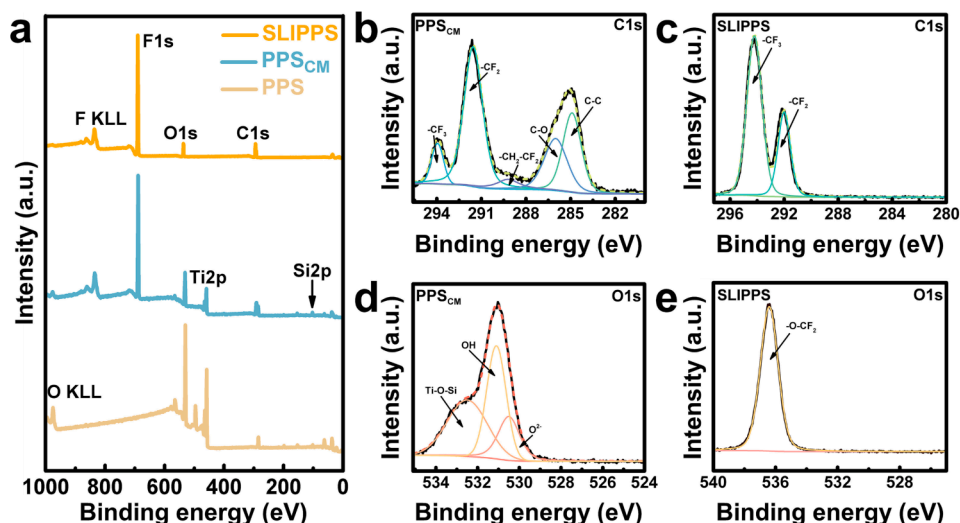


Fig. 4. (a) XPS spectra of the PPS, PPS_{CM} and SLIPPS. XPS spectrum deconvolution for the C1s peak of the (b) PPS_{CM} and (c) SLIPPS. XPS spectrum deconvolution for the O1s peak of the (d) PPS_{CM} and (e) SLIPPS.

PPS_{CM}, the $\theta_{A,r}$ values are 139.1° and 162.2°, respectively, while the $\theta_{B,r}$ values are appropriate 0° for both the PPS_{CM} and PPS_{CM}, which indicates the ultra-high wettability for the lubricant on these two surfaces. Besides, the relative surface roughness R is always larger than 1, for a porous surface [43,59]. Thus, for both the SLIPS and SLIPPS, the Eq. 1 and Eq. 2 are satisfied simultaneously, which means that in a static aqueous environment, the PFPE lubricant can be infused and retained spontaneously in the porous surface.

However, the working conditions for the practical applications are harsh, which will cause and further accelerate the loss of the PFPE lubricant. Therefore, in order to investigate the stability of the fabricated SLIPS and SLIPPS, the accelerated lubricant loss tests, such as the high-rate centrifugation and scouring, were performed to mimic the out-of-plane and in-plane forces of the water, respectively. In the high-rate centrifugation test, the SLIPS and SLIPPS samples were placed in the centrifuge for 1 min under different spin rates (0–12000 rpm), and the lubricated surface faced outwards along the direction of the centrifugal force to exert the out-of-plane force on the lubricant. After the centrifugation, the water Sliding Angle (SA), which is the tilted angle of the sample surface to initiate the sliding of the water droplet, were measured to ascertain the deterioration of the surface lubricity, as shown in Fig. 5a. Both the SAs of SLIPS and SLIPPS samples increase after the spin rate increases from 0 rpm to 12000 rpm, which indicates the loss of the lubricant and the reduction of the surface lubricity. However, under 9000 rpm, the SA of the SLIPPS keeps at about 5°, and even at 12000 rpm, the SA of SLIPPS samples remains below 10°. On the contrary, the SA of the SLIPS sample increases with the increase of spin rate, reaching $15.9^\circ \pm 1.6^\circ$ at 9000 rpm, and even worse, it reaches $23.9^\circ \pm 3.9^\circ$ at 12000 rpm. Based on the above results, apparently, the SLIPPS possesses the better capability to preserve the lubricant than that of the SLIPS, when being applied with an out-of-plane force. To quantitatively estimate the retained lubricant after the high-rate centrifugation, the weight loss of each sample was measured, as shown in Fig. S3 and Table S1. For the both two samples, the weight loss increases dramatically when the spin rate rises from 0 to 3000 rpm, due to the removal of the extra lubricant on the top of the sample surface. Then, the weight loss keeps increasing slightly when the spin rate further rises. Notably, no matter what the spin rate is, the residual lubricant mass of the SLIPPS is almost twice as much as that of the SLIPS. Besides, the distribution of the lubricant on the sample surface can be visualized to further analyze the capability of the sample to preserve the lubricant after a high-rate centrifugation [66,67]. Based on the micro-FTIR spectrum mapping results after a 12000-rpm centrifugation for 1 min in

Fig. S3a-b, the SLIPPS sample with a uniform red color evenly preserves more lubricant across its surface than that of the SLIPS sample, indicating that the firm mechanical interlock between the nano-prickles can effectively retain more lubricant. The both residual lubricant mass and micro-FTIR spectrum mapping results confirm that the SLIPPS with additional nano-prickles can preserve more lubricant under the out-of-plane force loading, which can be the reason why the SLIPPS still possesses a low SA even after a 12000-rpm centrifugation test. As a comparison, the SA of the pristine TA2 sample with the lubricant was also measured after being centrifuged, and a 3000-rpm spin rate can already result in a maximum SA (90°), as shown in Fig. S4, which indicates that there was almost no lubricant left on its surface at that time. Furthermore, to generally examine the anti-biofouling performance after the centrifugation test (12000 rpm, 1 min), both the SLIPS and SLIPPS samples were immersed in a suspension of a typical biofouling micro-organism (*Chlorella*) for 7 days. The corresponding Confocal Laser Scanning Microscopy (CLSM) images are shown in Fig. S5, the *Chlorella* coverage on the SLIPPS is only one fifth of that on the SLIPS, which indicates that the SLIPPS has the more outstanding capability to prevent from the adhesion of the *Chlorella*. As a summary, enhanced by the hierarchical nanostructure, especially the mechanical interlock provided by the nano-prickles, the SLIPPS has the superior capabilities to preserve the lubricant, keep the super-lubricity and resist the biofouling, when being applied with the out-of-plane force of the water.

On the other hand, in practical marine applications, the relative shear between the water flow and the material surface can generate the in-plane shear force to wash the lubricant away. Therefore, to explore the in-plane shear force influence on the stability of the anti-biofouling coating, the scouring test was performed. In the scouring test, the SLIPS and SLIPPS samples were placed in the water with different scouring speeds for 15 min, and then the SAs of SLIPS and SLIPPS was measured. As shown in Fig. 5b, the SA of SLIPPS remains below 4° after the water scouring with all different speeds, while the SA of SLIPS rises dramatically up to 10° with the increase of the scouring speed. Combined the results of this test and the high-rate centrifugation test, it can be concluded that the SLIPPS has a better stability of the super-lubricity than that of the SLIPS, undergoing the loading of either the out-of-plane force or the in-plane force.

On the basis of these results, the reasons for the SLIPPS with a better stability of the super-lubricity could be discussed as below. Obviously, the hierarchical nanostructure of the SLIPPS consists of micro/nanopores and nano-prickles can retain more lubricant. The micro/nanopores offer sufficient space to store the lubricant, and the tiny prickles

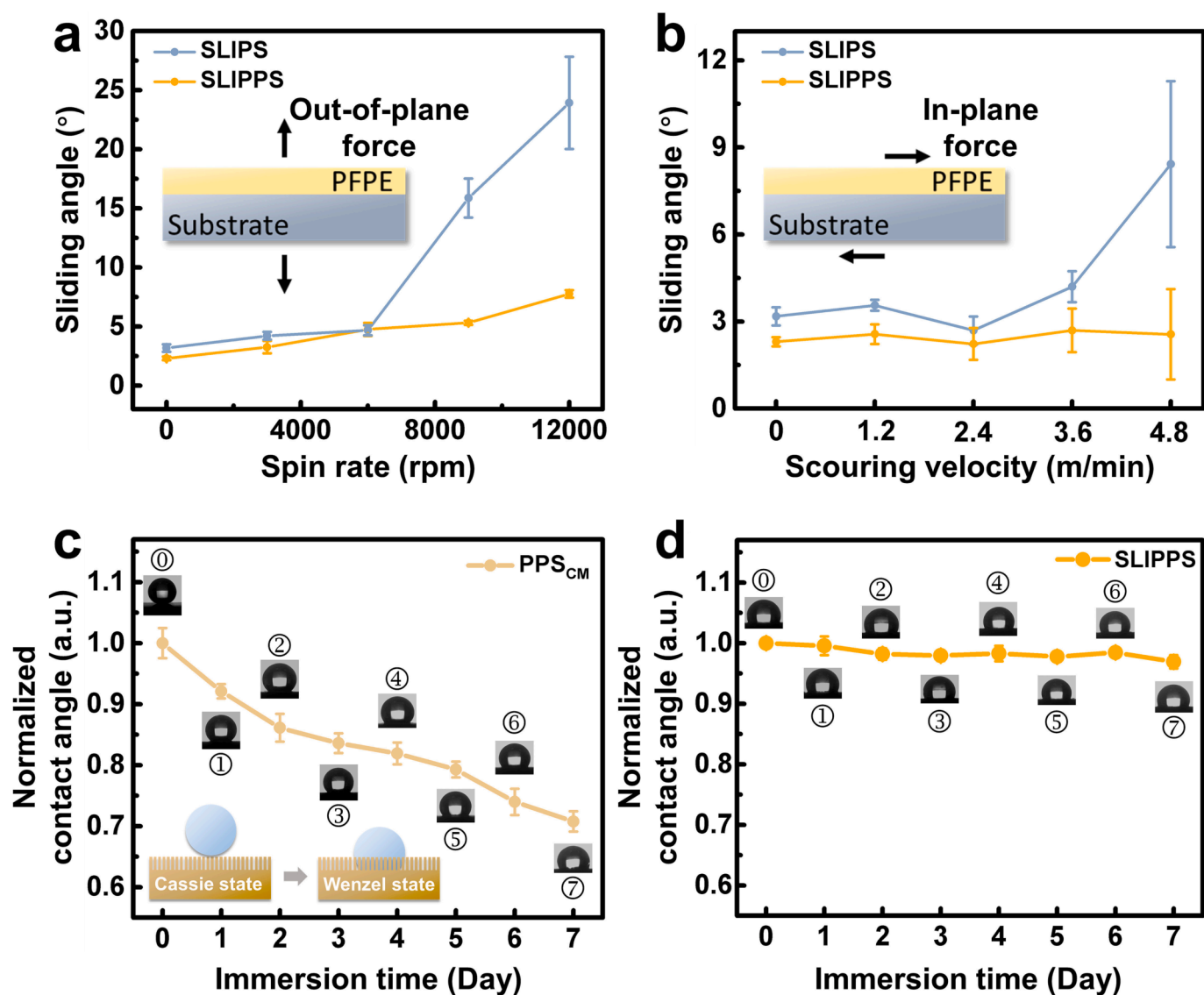


Fig. 5. Characterizations of the stability and durability of the anti-biofouling coatings. SAs of the SLIPS and SLIPPS after the (a) high-rate centrifugation and (b) scouring tests, respectively. Normalized CAs of the (c) PPS_{CM} and (d) SLIPPS after being immersed for different days.

on/inside the pores can provide the tremendous interface area and firm mechanical interlock with the lubricant. In contrast, the lubricant on the SLIPS is more easily washed away, and its smooth micro/nano-pores cannot offer a sufficient resistance against the loss of the lubricant for a long-time. Therefore, with this comparison between the SLIPS and SLIPPS, the additional nano-prickles in the SLIPPS plays a very important role to improve the stability of the super-lubricity and provide a stable anti-biofouling capability.

In general, the durability of the anti-biofouling coating is the key issue for its long-term service life in marine applications. To investigate the durability of the super-lubricated surface and super-hydrophobic surface, the CAs of the PPS_{CM} and SLIPPS was characterized by a 7-day immersion experiment with seawater. Since the super-hydrophobic PPS_{CM} and super-lubricated SLIPPS are two different types of surfaces with different anti-biofouling mechanisms, the direct comparison of the absolute CA values is not so meaningful. However, because the CA variation can reflect the change of the surface status, the normalized CAs are shown in Fig. 5c-d, which are divided by the initial CA of the corresponding sample before the immersion test. The initial CA of the super-hydrophobic PPS_{CM} is $162.2^\circ \pm 4.0^\circ$ and decreases down to 70.8° after the 7-day immersion, which indicates that the

wetting state changed from the Cassie state to the Wenzel state. This result shows that due to the detachment or dissolution of the trapped air, the water invaded into the surface nanostructure, and the surface status of the PPS_{CM} was profoundly modified. To further ascertain the influence of the immersion test on the surface status of the PPS_{CM}, the FTIR (Fourier Transform Infrared spectroscopy) tests were conducted with the PPS_{CM} sample before and after the immersion test, respectively. As show in Fig. S6a, before the immersion test, the major absorption peaks of the functional groups in the hydrophobic molecules are at 966 cm^{-1} , 1310 cm^{-1} and 1400 cm^{-1} , which correspond to the frequency of the Si-O-Si, the asymmetric stretching frequency of the CF₂ and the frequency of the CH₂ [68–70]. Nevertheless, after the 7-day immersion test, all the aforementioned absorption peaks disappear, which indicates the destruction of the hydrophobic organic layer on the PPS_{CM} and the possible deterioration of the anti-biofouling performance. On the other hand, for the super-lubricated SLIPPS, the initial CA is $117.0^\circ \pm 0.7^\circ$, and it only reduces down to 96.9° after the immersion test, as displayed in Fig. 5d. Besides, according to the comparison of the FTIR spectra before and after the 7-day immersion test in Fig. S6b, the absorption peaks are the same in the two spectra, indicating that the lubricant still covers on the samples surface after the immersion test. These results

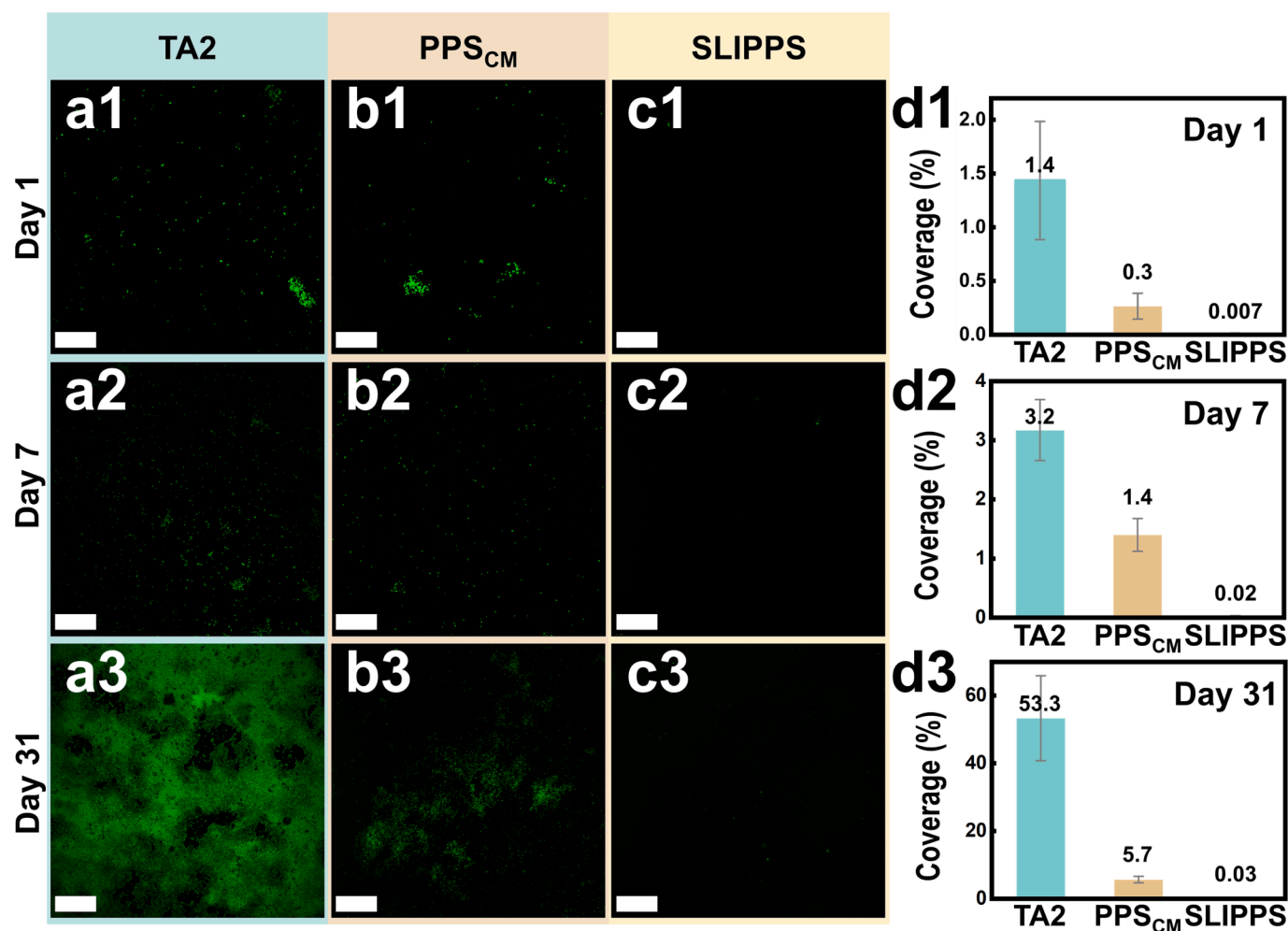


Fig. 6. CLSM images of the Chlorella covering on the (a1 - a3) TA2, (b1 - b3) PPS_{CM} and (c1-c3) SLIPPS after being immersed in a suspension of Chlorella for 1, 7 and 31 days, respectively. Statistical results of the Chlorella coverage on (d1) TA2, (d2) PPS_{CM} and (d3) SLIPPS after being immersed in a suspension of Chlorella for 1, 7 and 31 days, respectively. All the scale bars are 200 μm .

imply that for the SLIPPS, there is no obvious change of its surface status, during the 7-day immersion test with seawater. Apparently, the durability of the super-lubricated SLIPPS is much more outstanding than that of the super-hydrophobic PPS_{CM}, when being immersed with seawater. Then, to further examine the service performance of the SLIPPS, its corrosion resistance and biofouling resistance are presented in the following sections.

3.4. Corrosion resistance of anti-biofouling coatings

Besides the biofouling resistance, the corrosion resistance is another crucial property for materials in marine environment. Hence, to investigate the corrosion resistance, different samples were characterized to obtain the potentiodynamic curves, which are plotted in Fig. S7. And, the potentiodynamic curve parameters are shown in Table 1. In general, a lower corrosion current density (i_{corr}) means fewer transferred charges and a lower corrosion dynamic rate, and the more positive the corrosion potential (E_{corr}) of the sample, the lower thermodynamical tendency to

corrode [71]. Compared to the pristine TA2 sample, the E_{corr} of the PPS_{CM} is much positive, and the i_{corr} of the PPS_{CM} is 1 order of magnitude lower. The trapped air layer on the PPS_{CM} plays a significant role in preventing the corrosive medium from contacting the TA2 surface. On the other hand, due to the stable and durable existence of the lubricant on the SLIPPS, the corrosive medium is completely isolated from the material surface, and the E_{corr} of the SLIPPS is only -17.2 mV, which is the most positive among the samples. The i_{corr} of the SLIPPS is 2 orders of magnitude lower than that of TA2. These results indicate that the SLIPPS has the lowest thermodynamical tendency to corrode and the smallest corrosion dynamic rate. Hence, the existence of the electrochemically inert lubricant layer in the SLIPPS can significantly increase the resistance to the corrosion and slow down the corrosion process.

3.5. Biofouling resistance of anti-biofouling coatings

To evaluate the anti-biofouling performance of the as-fabricated samples, two typical biofouling microorganisms were selected and cultured with them. Chlorella is one of the most common biofouling microorganisms and widely cultured in labs, while diatom is the most diverse group of the unicellular eukaryotes in oceans and other aquatic systems [72].

In the anti-biofouling tests with the Chlorella, the TA2, PPS_{CM} and SLIPPS were immersed into a culture media with Chlorella for 1 day, 7 days and 31 days, respectively. Then, the coverage results of the

Table 1
Electrochemical parameters of the TA2, PPS_{CM} and SLIPPS.

Samples	E_{corr} (V)	i_{corr} ($\text{A}\cdot\text{cm}^{-2}$)
TA2	-0.439	3.36×10^{-7}
PPS _{CM}	-0.221	4.37×10^{-8}
SLIPPS	-0.0172	1.11×10^{-9}

Chlorella cells on the sample surfaces were characterized in the CLSM images, as shown in Fig. 6a-c, and the corresponding statistical numbers are summarized in Fig. 6d. For the pristine TA2 sample without any surface treatment, the Chlorella cells sporadically cover a 1.4% surface within only 1 day. After the continuous immersion for 7 and 31 days, the coverage percentages of the Chlorella cells on the TA2 substantially increases up to 3.2% and 53.3%, respectively. Finally, the Chlorella cells form a denser biofilm, which can be considered as a severe biofouling result. For the super-hydrophobic PPS_{CM}, the coverage percentages of the Chlorella cells within 1, 7 and 31 days are 0.3%, 1.4% and 5.7%, respectively. Due to its super-hydrophobicity and a trapped air layer isolating the water, the PPS_{CM} possesses a better anti-biofouling performance than that of the pristine TA2. However, with the trapped air layer gradually detaching or dissolving, the water invades into the surface nanostructure, and the isolation barrier between the Chlorella cells and material surface disappears. Then, due to the destruction of the hydrophobic organic layer, the PPS_{CM} loses its super-hydrophobicity (Fig. 5c), and the Chlorella cells attach much more easily and firmly [48]. In Fig. 6b3, considerable Chlorella cells can be observed on the PPS_{CM} surface, which implies the development of the biofouling.

On the contrary, the coverage percentage of the Chlorella cells on the SLIPPS is only 0.007% for a 1-day immersion. Even for a 31-day immersion, the coverage percentage of the Chlorella cells on the SLIPPS is still only 0.03%. This coverage percentage is 3 and 2 orders of magnitude less than those on the TA2 and PPS_{CM}, respectively. It can be concluded that the SLIPPS keeps its outstanding anti-biofouling performance after the aforementioned 31-day immersion. And outstanding anti-biofouling performance is attributed to the excellent stability and durability of the lubricant layer on the SLIPPS, which can offer a super-lubricated surface and isolation barrier for a long time period to prevent

the attachment of the Chlorella cells.

On the other hand, *Phaeodactylum tricornutum* was adopted as the second microorganism in the anti-biofouling tests to represent the diatoms, since it is the model organism of the diatoms. As shown in Fig. 7a, after being immersed for 1 day, a few *Phaeodactylum tricornutum* cells attach on the pristine TA2 surface with a coverage percentage of 0.1%. As the immersion time grows, the coverage percentage increases significantly up to 9.1% and 23.2%, when the TA2 sample is immersed for 7 and 31 days, respectively. Similar to the anti-biofouling tests with the Chlorella, the PPS_{CM} can provide a short-term protection before the trapped air layer detaching or dissolving. It also can be found that the coverage percentages of the *Phaeodactylum tricornutum* cells on the PPS_{CM} are 0.1%, 0.9%, and 17.2% for 1 day, 7 days and 31 days, respectively.

In contrast, the biofouling resistance to the *Phaeodactylum tricornutum* cells of the SLIPPS is outstanding as well as that to the Chlorella cells. The coverage percentages of the *Phaeodactylum tricornutum* cells on the SLIPPS are 0.009%, 0.02% and 0.06% after being immersed for 1 day, 7 days, and 31 days, respectively. Obviously, the SLIPPS can effectively inhibit the attachment of the *Phaeodactylum tricornutum* cells, and the coverage percentage of the *Phaeodactylum tricornutum* cells on the SLIPPS is 2 orders of magnitude less than those on the TA2 and PPS_{CM} for 31 days. Based on these anti-biofouling tests with two different common biofouling microorganisms, it can be concluded that the SLIPPS with its excellent long-term stability and durability possesses outstanding anti-biofouling performance and can effectively prevent the attachment of the biofouling microorganisms. Additionally, the corresponding anti-biofouling performance of the SLIPPS is presented in Fig. S8, which is the second-best to that of the SLIPPS.

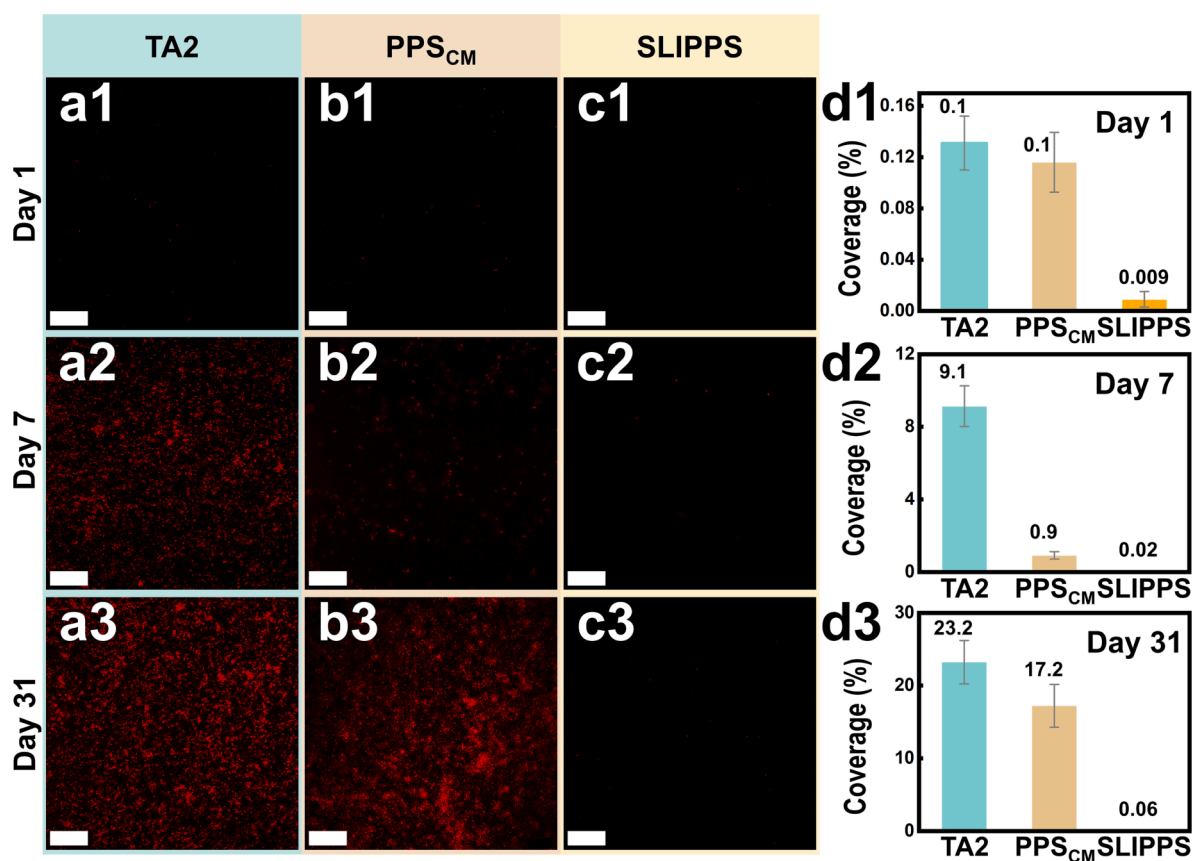


Fig. 7. CLSM images of the *Phaeodactylum tricornutum* covering on the (a1 - a3) TA2, (b1 - b3) PPS_{CM} and (c1-c3) SLIPPS after being immersed in a suspension of *Phaeodactylum tricornutum* for 1, 7 and 31 days, respectively. Statistical results of the *Phaeodactylum tricornutum* coverage on (d1) TA2, (d2) PPS_{CM} and (d3) SLIPPS after being immersed in a suspension of *Phaeodactylum tricornutum* for 1, 7 and 31 days, respectively. All the scale bars are 200 μ m.

3.6. Anti-biofouling mechanism of coatings

According to the anti-biofouling test results of the TA2, PPS_{CM} and SLIPPS samples, the schematic diagrams of the possible anti-biofouling mechanism are shown in Fig. 8. It is believed that titanium alloys are highly biocompatible and attractive to marine organisms, which rises a serious of biofouling problems. Due to its super-hydrophobic property, the PPS_{CM} surface has a trapped air layer when being immersed into the water, which can ensure an anti-biofouling barrier for a short term (Fig. 8a). However, for a long-term immersion in water, the trapped air layer will detach or dissolve, and the super-hydrophobic property will eventually disappear by the destruction of the hydrophobic organic layer on the PPS_{CM}, as shown in Fig. 5c and Fig. S6a. Furthermore, the changes of surface chemical composition were characterized by XPS. The results show that the positive ions (e.g.: Ca²⁺, Na⁺ and so on) in the seawater can accumulate on the surface of the PPS_{CM} sample (Fig. S9a). This surface with positive ions is more attractive to the biofouling microorganisms via the Coulomb force, since the biofouling microorganisms and the SLIPPS are both negatively charged (Fig. S10), which can finally result in a deterioration of the anti-biofouling performance [73,74]. On contrary, due to its outstanding stability and durability, the SLIPPS can remain the initial state and keep the positive ions away (Fig. S9b), during the immersion test.

Without doubt, the stable and durable existence of the lubricant is the most important factor for the excellent anti-biofouling performance of the SLIPPS. In this work, the novel in-situ fabricated hierarchical nanostructure on titanium alloy can provide the sufficient storage for the lubricant and the enhanced mechanical interlock to ensure the long-term stability and durability of the SLIPPS. This stable and durable lubricant layer can offer multiple mechanisms to protect the vulnerable

titanium alloy surface from biofouling. Firstly, the PFPE, as the lubricant, is a non-volatile liquid and not immiscible with both aqueous and hydrocarbon phases [32]. Unlike the oil–water interface containing alkanes may attractive to biofouling microorganisms [75], the PFPE is a widely adopted lubricant in super-lubricated surfaces for anti-biofouling applications [37,39,45,76–78]. This inert lubricant layer can physically isolate the substrate material from the biofouling microorganisms. Secondly, the lubricant layer can mimic the epidermal secretion on the skins of the cetaceans to form a slippery surface, which is changeable, shapeless and mobile. This kind of surfaces is difficult for the biofouling microorganisms to attach tightly. Normally, the biofouling microorganisms prefer solid and stationary surfaces to settle down, because this type of surfaces can offer firm and permanent connections with the biofouling creatures [39]. In contrast, the slippery lubricant layer can inhibit the formation of any connect between the surface and microorganism cells. Besides, due to the highly slippery surface of the sample, if any microorganism cell has accidentally attached, it can be easily removed by a tiny seawater flow. Last but not least, the abundant fluorine atoms in the PFPE are negatively charged [79–81], which can repel the biofouling microorganism cells by the repulsive Coulomb force [82], because these cells are always negatively charged as well (Fig. S10). This repulsive force can reduce the chance that the biofouling microorganism cells get close to the surface and further prevent from their attachment. Eventually, the synergistic effect of the aforementioned mechanisms ensures the remarkable anti-biofouling performance of the SLIPPS on the titanium alloy substrate.

4. Conclusion

In this work, a hierarchically nanostructured super-lubricated

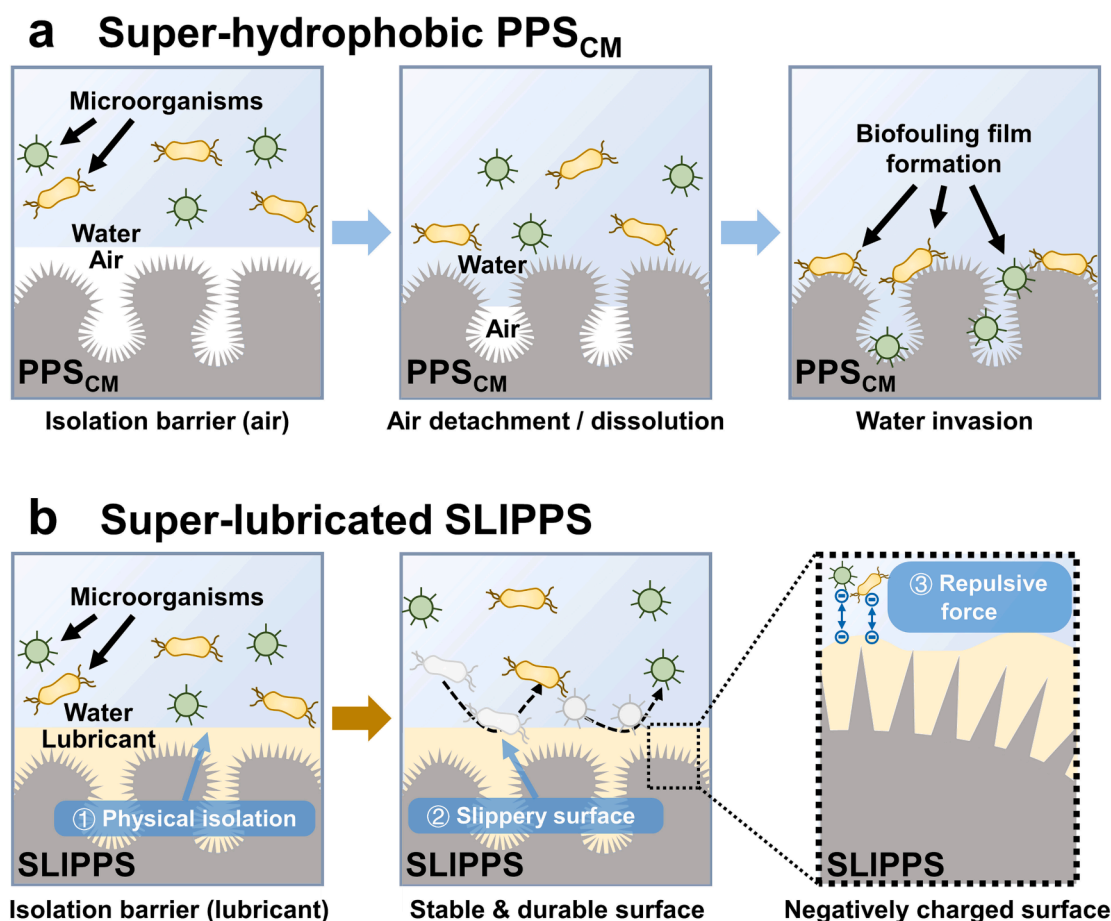


Fig. 8. Schematic diagram of the anti-biofouling mechanism of the SLIPPS.

surface on a titanium alloy substrate is successfully developed via the micro-arc oxidation and hydrothermal processes. The hierarchical structure consists of two parts: micro/nano-pores and nano-prickles inside/on the porous surface. The experimental results confirm that the micro/nano-pores can offer sufficient space to store the lubricant, and the tiny prickles on/inside the pores can provide the firm mechanical interlock with the lubricant. Based on the high-rate centrifugation, scouring and long-time immersion tests, it is proved that the SLIPPS has an excellent capability to stably and durably preserve the lubricant and maintain a super-lubricated surface, when being exerted by either an out-of-plane force or an in-plane force. As an electrochemically inert material, the stable and durable lubricant layer can significantly improve the corrosion resistance of TA2. The E_{corr} of the SLIPPS is only -17.2 mV, which is much more positive than that of the TA2 (-439 mV). Meanwhile, the I_{corr} of the SLIPPS is 2 orders of magnitude less than that of TA2. These results indicate that the SLIPPS possesses an outstanding corrosion resistance, which can ensure a long-time service life in the corrosive seawater.

The anti-biofouling performance was evaluated by the long-term immersion tests with typical biofouling microorganisms, like *Chlorella* and *Phaeodactylum tricornutum*. The coverage percentages of the *Chlorella* and *Phaeodactylum tricornutum* cells on the SLIPPS is only 0.03% and 0.06% after being immersed for 31 days, respectively, which is 3 and 2 orders of magnitude less than that on the TA2, respectively. These results convincingly demonstrate that the SLIPPS can almost completely prevent from the attachment of the biofouling microorganisms for quite a long time. This excellent anti-biofouling performance can result from the synergistic effect of multiple mechanisms, including the physical isolation, slippery surface and repulsive Coulomb force. This work demonstrates that enhanced by the hierarchical nano-structures, this novel SLIPPS has the remarkable stability and durability, anti-biofouling performance and corrosion resistance. Therefore, this in-situ fabricated, low-cost and super-lubricated coating on the titanium alloy surface could pave the way for the large-scale applications of titanium alloys in marine engineering.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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Appendix A. Supplementary data

Photos of the fabricated samples, SEM image of the cross-section of the PPS, Micro-FTIR spectrum mapping results of the SLIPS and SLIPPS samples after a 12000-rpm centrifugation for 1 min, FTIR spectrum of the lubricant (PFPE), Residual masses of the lubricant after the 12000-rpm centrifugation for the SLIPS and SLIPPS samples, A 90° SA of the pristine TA2 after a centrifugation with a 3000-rpm spin rate, CLSM images of the *Chlorella* covering on the SLIPS and SLIPPS after being

immersed in a suspension of *Chlorella* for 7 days, FTIR spectra of the PPS_{CM} and SLIPPS before and after the immersion test for 7 days, Potentiodynamic curves of the TA2, PPS_{CM} and SLIPPS, CLSM images of the *Chlorella* and *Phaeodactylum tricornutum* covering on the SLIPS after being immersed in a suspension of *Chlorella* and *Phaeodactylum tricornutum* for 1, 7 and 31 days, respectively, Statistical results of the *Chlorella* and *Phaeodactylum tricornutum* coverages on the SLIPS after being immersed in a suspension of *Chlorella* and *Phaeodactylum tricornutum* for 1, 7 and 31 days, respectively, XPS spectra of the PPS_{CM} and SLIPPS before and after the immersion test for 7 days, Zeta potentials of *Chlorella*, *Phaeodactylum tricornutum* and SLIPPS (PDF). Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cej.2023.142389>.

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